

EZ-PD™ BCR-PLUS/BCR-LITE Host Processor Interface Specification

CYPD3176/CYPD3178

About this document

Scope and purpose

Scope and purpose

This document defines the Host Processor Interface Specification for the devices of the EZ-PD BCR-PLUS/BCR-LITE product family which includes CYPD3176 and CYPD3178 devices. The document summarizes the registers which can be accessed from the system controller.

Intended audience

This document is intended for the users of EZ-PD BCR-PLUS/BCR-LITE devices to access the internal registers using the system controller through Host Processor Interface.

Document conventions

Table 1 Conventions

Convention	Explanation
Bold	Emphasizes heading levels, column headings, table and figure captions, screen names, windows, dialog boxes, menus and sub-menus
Italics	Denotes variable(s) and reference(s)
Courier New	Denotes APIs, functions, interrupt handlers, events, data types, error handlers, file/folder names, directories, command line inputs, code snippets
>	Indicates that a cascading sub-menu opens when you select a menu item

About this document

Abbreviations and definitions

Table 2 Abbreviations

Abbreviation	Definition
AFC	Adaptive fast charging
BCR	Barrel connector replacement
CC	Configuration channel
CCG	Cable controller generation
DFP	Downstream facing port
DNP	Do not populate
EC	Embedded controller
EMCA	Electronically marked cable assembly
ESD	Electrostatic discharge
EVK	Evaluation kit
FET	Field-effect transistor
GPIO	General-purpose input/ output
IC	Integrated circuit
I ² C	Inter-integrated circuit
LED	Light-emitting diode
NA	Not applicable
OVP	Over-voltage protection
PA	Power adapter
PD	Power delivery
PDO	Power data object
PFET	P-channel field effect transistor
PSoC™	Programmable system-on-chip
QC®	Qualcomm quick charge
SDK	Software development kit
SoC	System-on-chip
UFP	Upstream facing port
USB	Universal serial bus
USB-PD	Universal serial bus power delivery
UVP	Under voltage protection
OVP	Over-voltage protection
OCP	Over current protection
HPI	Host processor interface

Table of contents

About this document.....	1
Table of contents.....	3
1 Introduction	5
1.1 Modes of operation	5
1.2 Introduction to HPI.....	5
1.3 HPI capabilities.....	6
2 I2C transport	7
2.1 I2C Write to BCR-PLUS/BCR-LITE.....	7
2.2 I2C Read from BCR-PLUS/BCR-LITE.....	7
2.3 Interrupt (Alert) pin from BCR-PLUS/BCR-LITE.....	8
3 HPI protocol	9
3.1 Types of registers	9
3.2 Reading status from BCR-PLUS/BCR-LITE (Status Registers).....	9
3.3 Command – Response Model (Command/Response Registers)	9
3.4 Asynchronous PD messages and events (Response registers)	10
3.5 Flow diagrams	10
3.6 Data memory accesses.....	10
3.7 Applicability of the HPI registers	11
3.8 Summary of registers	12
3.9 DEVICE_MODE	13
3.10 SILICON_ID	13
3.11 INTERRUPT	14
3.12 PD_STATUS.....	14
3.13 TYPE_C_STATUS.....	15
3.14 BUS_VOLTAGE	16
3.15 BUS_CURRENT	16
3.16 CURRENT_PDO	17
3.17 CURRENT_RDO	17
3.18 EVENT_STATUS	18
3.19 CHARGING_MODE_STATUS	18
4 Command registers	19
4.1 RESET	19
4.2 EVENT_MASK	19
4.3 DM_CONTROL (Send PD Data Message)	21
4.4 SELECT_SINK_PDO	22
4.5 PD_CONTROL (Send PD control message)	23
4.6 SINK_RDO_REQUEST	24
4.6.1 PPS Request	25
4.7 FAULT_PIN_MASK	25
4.8 OVP_CONFIG.....	26
4.9 OCP_CONFIG	27
4.10 OTP_CONFIG.....	28
4.11 SNK_FET_DISABLE	29
4.12 SNK_FET_CONTROL.....	29
4.13 SAFE_PWR_CONTROL	30
4.14 CHARGING_MODE_DISABLE	30
5 Response registers	31

Table of contents

5.1	DEV_RESPONSE	31
5.2	PD_RESPONSE.....	31
5.3	Response codes.....	32
5.4	Data Memory Regions	37
Revision history.....		38
Disclaimer.....		39

Introduction

1 Introduction

EZ-PD™ Barrel Connector Replacement Plus (BCR-PLUS)/ EZ-PD™ Barrel Connector Replacement Lite (BCR-LITE) is an easy-to-use Type-C and Power Delivery controller for electronic devices that consume power over the Type-C port. With a few resistor settings, you can customize EZ-PD™ BCR-PLUS/ EZ-PD™ BCR-LITE to match your design requirement, without having to write firmware or load configuration options over a protocol interface.

Both CYPD3176-24LQXQ (BCR-PLUS) and CYPD3178-LQXQ(BCR-LITE) ships with pre-programmed firmware with functionality as documented in the datasheet. An additional configuration table is an option available for customers who prefer further customization of the BCR-PLUS solution using the EZ-PD™ configuration utility tool to fit the needs of their end- application.

In addition to the resistor settings, the CYPD3176-24LQXQ(BCR-PLUS)/CYPD3178-24LQXQ(BCR-LITE) device can also be connected to a microcontroller or processor that wants more information about the current state of the Type-C port or wants to dynamically change the power settings of the device.

The Host Processor Interface protocol over I2C is used for just this purpose.

1.1 Modes of operation

The EZ-PD™ BCR-PLUS/ EZ-PD™ BCR-LITE device can operate in one of the following two modes

1. Standalone mode
2. Host Processor Interface (HPI) mode

In the standalone mode, the EZ-PD™ BCR-PLUS/ EZ-PD™ BCR-LITE device uses the voltage and current settings specified using resistor dividers on its four configuration pins.

In HPI mode, the EZ-PD™ BCR-PLUS/ EZ-PD™ BCR-LITE device is controlled by an external I2C master, which specifies the voltage and current options it requires at any point.

The Host Processor Interface is Active in both modes, but certain write-able registers are only available in the HPI mode.

1.2 Introduction to HPI

The Host Processor Interface is a register-based protocol over an I2C slave interface. Frequencies up to 400kHz are supported and an interrupt line is provided as an alert indicator.

The connection between a system controller and EZ-PD™ BCR-PLUS/ EZ-PD™ BCR-LITE is as shown in [Figure 1](#).

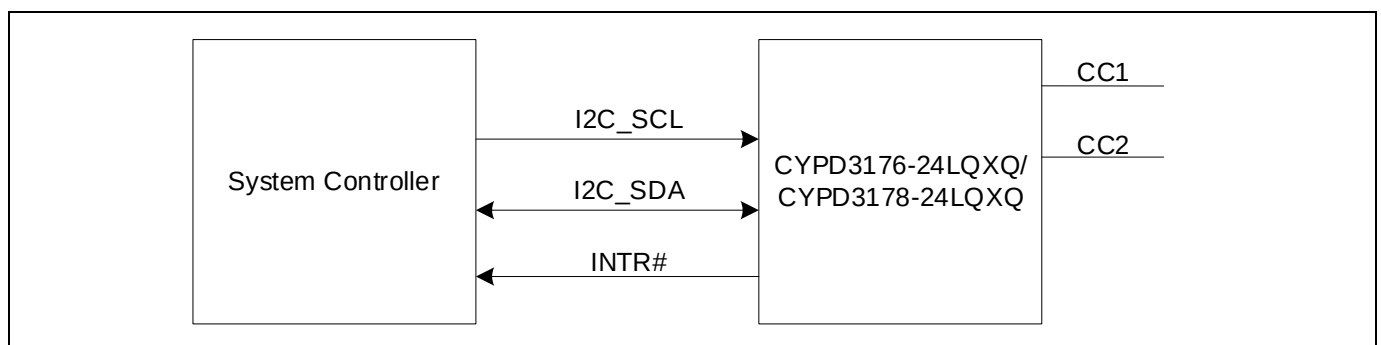


Figure 1 Connections between system controller and EZ-PD™ BCR-PLUS/ EZ-PD™ BCR-LITE device

Introduction

1.3 HPI capabilities

The main capabilities provided by EZ-PD™ BCR-PLUS/ EZ-PD™ BCR-LITE to the system controller using HPI include:

- Firmware version identification
- Firmware update capability (only applicable to EZ-PD™ BCR-PLUS device)
- Reporting of Type-C(applicable for both devices) and USB-PD connection status(only applicable to EZ-PD™ BCR-PLUS device)
- Interrupt-based event reporting when connection status changes
- Control of USB-PD power profiles (only applicable to EZ-PD™ BCR-PLUS device)
- Ability to send and receive USB-PD Vendor Defined Messages (VDMs) (only applicable to EZ-PD™ BCR-PLUS device)
- Ability to initiate USB-PD control messages including resets, role swaps etc. (only applicable to the EZ-PD™ BCR-PLUS device)

EZ-PD™ BCR-PLUS/ EZ-PD™ BCR-LITE provides these functionalities using commands, responses, events, and asynchronous messages modeled as registers over the I2C slave interface.

A set of read-only **Status Registers** report the current state of the Type-C port and Power Delivery contract.

Command Registers provide commands to perform Type-C and PD interface related tasks. The system controller can send commands to EZ-PD™ BCR-PLUS/ EZ-PD™ BCR-LITE by writing to these registers. EZ-PD™ BCR-PLUS/ EZ-PD™ BCR-LITE processes these commands and responds back through the [DEV_RESPONSE](#) or [PD_RESPONSE](#) registers.

EZ-PD™ BCR-PLUS/EZ-PD™ BCR-LITE also notifies the system controller about Type-C and PD specific events and messages through the [PD_RESPONSE](#) register.

The INTR line is used to notify the system controller about response, event, and asynchronous messages.

In addition to the status, command, and response registers; the HPI address map includes a 256-byte Data Memory space. This space is used as a buffer area to exchange data associated with the commands, responses, and events.

I2C transport

2 I2C transport

The HPI uses I2C as a transport mechanism to read and write to registers inside the device. The following sections describe how reads and writes are done.

2.1 I2C Write to BCR-PLUS/BCR-LITE

I2C writes begin with a Start bit and the write preamble of 7-bit device address and the Write bit (0). EZ-PD™ BCR-PLUS/ EZ-PD™ BCR-LITE responds back with a 1-bit ACK/NAK response.

EZ-PD™ BCR-PLUS/ EZ-PD™ BCR-LITE uses 0x08 as the 7-bit device address.

The first two bytes following a write preamble is the address of the 16-bit register. The least significant byte (LSB) of a register address is transferred first followed by the most significant byte. All bytes following the address bytes are the register data.

Figure 2 shows the bus sequence for a two-byte I2C write operation.

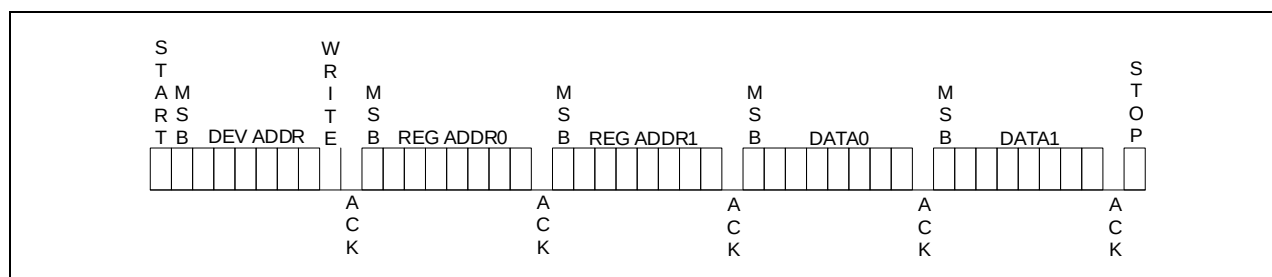


Figure 2 I2C write sequence

Writes to invalid register addresses are ignored.

2.2 I2C Read from BCR-PLUS/BCR-LITE

I2C reads begin with a Start bit and read preamble where the system controller writes the 16-bit Register address that it wants to read from.

The address is followed by another Start bit (i.e. restart) on the bus, instead of a Stop bit. The master then sends a read preamble with 7-bit device address and Read bit (1) and reads data from the device.

The maximum read size is 512 bytes. Reads beyond the valid address region will be NAK-ed by the BCR-PLUS/BCR-LITE device. Figure 3 shows the I2C transfer sequence for a two-byte read operation.

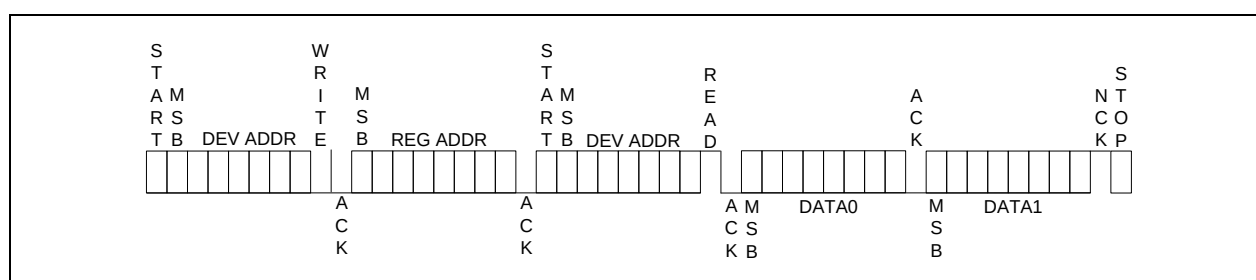


Figure 3 HPIv1 I2C read sequence

EZ-PD™ BCR-PLUS/ EZ-PD™ BCR-LITE will NAK the read preamble if the I2C master does not write a register address followed by Restart before reading register contents.

I2C transport

The I2C master on the system controller must support clock stretching. EZ-PD™ BCR-PLUS/ EZ-PD™ BCR-LITE will stretch the I2C clock in the following scenarios:

1. At the acknowledge (ACK/NAK) phase of the read or write preamble. EZ-PD™ BCR-PLUS/ EZ-PD™ BCR-LITE stretches the clock for 3~5 cycles before responding to the preamble sent by the I2C master.
2. EZ-PD™ BCR-PLUS/ EZ-PD™ BCR-LITE has an 8-byte FIFO to receive data sent by I2C master. The clock is stretched at the acknowledgement phase of every 4~8 data bytes as the FIFO gets filled up and the internal logic drains bytes to process them.

2.3 Interrupt (Alert) pin from BCR-PLUS/BCR-LITE

The INTR pin is an Active Low output I/O that EZ-PD™ BCR-PLUS/ EZ-PD™ BCR-LITE drives to notify the system controller of responses, events and asynchronous messages. The INTR is an open drain signal that can only be driven LOW. An external pull up (~5 kΩ) is required.

The cause of the interrupt can be detected by reading the interrupt status register (see [INTERRUPT](#)).

Two separate response registers are provided for general device events as well as events corresponding to the USB Type-C port. Separate interrupt status bits are provided for each of these event types.

The EZ-PD™ BCR-PLUS/ EZ-PD™ BCR-LITE device maintains an event queue. The INTR pin will be re-asserted until the system controller clears all events from the queue. The BCR-PLUS/BCR-LITE waits for 50 μs (after system controller clears the interrupt) before re-asserting INTR.

3 HPI protocol

3.1 Types of registers

EZ-PD™ BCR-PLUS/ EZ-PD™ BCR-LITE has four types of registers:

1. Read-only Status Registers
2. Write-only Command Registers
3. Read-only Response Registers
4. Read/Write Data Memory

3.2 Reading status from BCR-PLUS/BCR-LITE (Status Registers)

The Status Registers are read-only and contain the state of the device, the Type-C port, and the Power Delivery contract. A read of these registers will not trigger any assertion of INTR.

3.3 Command – Response Model (Command/Response Registers)

If the system controller wants to control any aspect of BCR-PLUS/ BCR-LITE, then it must write to one or more Command Registers.

The interaction model with these registers is as follows:

1. The system controller sends a command to BCR-PLUS/ BCR-LITE by writing to a Command Register
2. BCR-PLUS/BCR-LITE executes the command and sends a response code back to the system controller through one of the two Response Registers
3. If the response has any data associated with it, the data is returned through the data memory region. BCR-PLUS/BCR-LITE uses the INTR# pin to notify the system controller that a response is available in the register
4. The system controller then clears the interrupt by writing to the INTERRUPT register. The EZ-PD™ BCR-PLUS/ EZ-PD™ BCR-LITE then de-asserts the INTR pin

The response to a command is sent in either the [DEV_RESPONSE](#) register (for device-specific commands) or the [PD_RESPONSE](#) register (for port-specific commands). The [INTERRUPT](#) register contains separate status bits for each of these response types and the INTR pin will be asserted if at least one of these response registers contain a message.

The system controller can identify the response register to be serviced and can clear only those specific bits in the [INTERRUPT](#).

The EZ-PD™ BCR-PLUS/ EZ-PD™ BCR-LITE uses two queues to hold responses and other events to be sent to the system controller. These queues can hold a limited number of entries and can overflow if the system controller does not read out the data in time.

The EZ-PD™ BCR-PLUS/ EZ-PD™ BCR-LITE clears the contents of a specific Response register only after EC writes to the corresponding bit in the [INTERRUPT](#) register. If the system controller sends a new command before reading the outstanding response to a previous command, the response to new command is queued in the internal message queue.

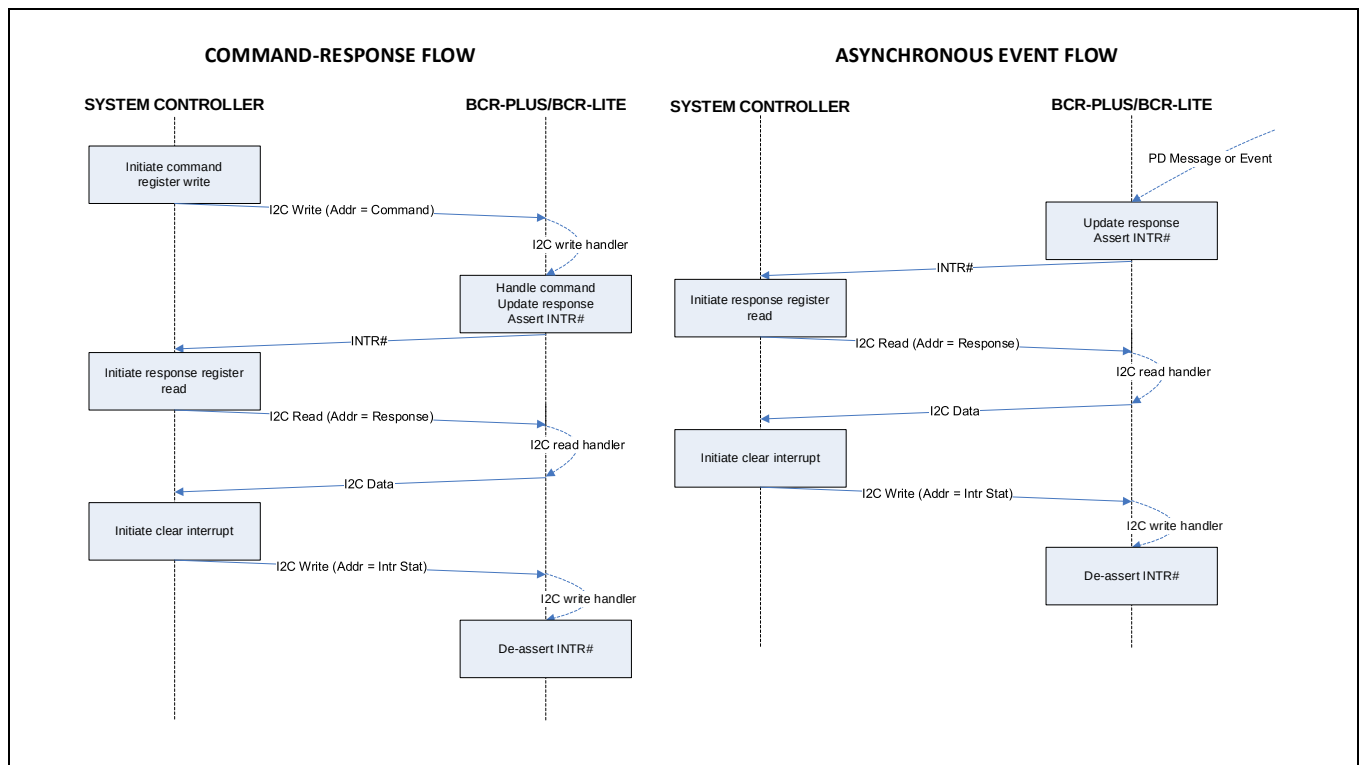
There can be a delay of up to 50 µs from the time the system controller clears the interrupt to the time the INTR signal is de-asserted. The system controller should take care to not sample the INTR signal in this period.

3.4 Asynchronous PD messages and events (Response registers)

In addition to responses to commands received, the PD_RESPONSE register is also used by EZ-PD™ BCR-PLUS/ EZ-PD™ BCR-LITE to notify the system controller about asynchronous USB Power Delivery messages. BCR-PLUS/BCR-LITE will assert the INTR pin when there is a new response or event stored in the response register. The system controller is expected to clear the interrupt after reading the response/event from the registers.

These asynchronous events can be enabled by writing to specific bits of the [EVENT_MASK](#) register.

3.5 Flow diagrams



3.6 Data memory accesses

The EZ-PD™ BCR-PLUS/ EZ-PD™ BCR-LITE supports separate read and write data memory regions (see [Data Memory](#))

Any data memory access operation (valid or invalid) will not result in a response or event generation. If the read operation crosses the data memory boundary, the EZ-PD™ BCR-PLUS/ EZ-PD™ BCR-LITE will NAK the transfer. If the write operation crosses the data memory boundary, the EZ-PD™ BCR-PLUS/ EZ-PD™ BCR-LITE drops further data bytes written by the system controller. No action is taken based on the write transaction and data memory is not updated in that situation.

HPI protocol

3.7 Applicability of the HPI registers

The following table details the HPI registers that are supported /not supported by the BCR-PLUS and BCR-LITE devices.

The following abbreviations are used:

- NS – Not Supported; shall not use, may not return useful information
- S- Supported; Shall be supported by the device

HPI register name	BCR-PLUS (CYPD3176)	BCR-LITE (CYPD3178)	Brief description
DEVICE_MODE	S	S	Indicates the active device mode
SILICON_ID	S	S	Read the Silicon ID of the device
INTERRUPT	S	S	Indicates the type of interrupt for which INTR pin was asserted
PD_STATUS	S	NS	Reports the Default PD configuration and the current state of the PD contract
TYPE_C_STATUS	S	S	Reports the status of the Type-C port
BUS_VOLTAGE	S	S	Report the live voltage on the VBUS
BUS_CURRENT	S	S	Report the live current on the VBUS
CURRENT_PDO	S	NS	Holds the active PDO requested
CURRENT_RDO	S	NS	Reports the Request Data Object (RDO)
EVENT_STATUS	S	S	Report the event status
CHARGING_MODE_STATUS	S	S	Reports the active Legacy Charging mode
RESET	S	S	Reset the device or the I2C block
EVENT_MASK	S	S	Enable the set of events
DM_CONTROL	S	NS	Send various Power Delivery Data Messages
SELECT_SINK_PDO	S	NS	Selectively enable or replace Sink Capabilities
PD_CONTROL	S	NS	Send Power Delivery Control messages
SINK_RDO_REQUEST	S	NS	Send a custom Request Data Object (RDO)
FAULT_PIN_MASK	S	S	Sets mask of fault conditions on the Fault pin
SNK_FET_DISABLE	S	NS	Disable SNK FET during USB-PD contract renegotiation
SNK_FET_CONTROL	S	S	Consumer path behavior control
SAFE_PWR_CONTROL	S	S	SAFE_PWR_EN feature control
OVP_CONFIG	S	S	Reconfigure over-voltage protection configuration
OCP_CONFIG	S	NS	Reconfigure over-current protection configuration
OTP_CONFIG	S	NS	Reconfigure over-temperature protection configuration
CHARGING_MODE_DISABLE	S	S	Disable the legacy charging mode support

HPI protocol

HPI register name	BCR-PLUS (CYPD3176)	BCR-LITE (CYPD3178)	Brief description
DEV_RESPONSE	S	S	Store device response
PD_RESPONSE	S	S	Store port response
Read Data Memory	S	S	Data Memory region for read
Write Data Memory	S	S	Data Memory region for write

3.8 Summary of registers

The following table lists the typical uses of HPI and the specific registers used to perform the action. Note that the list is not exhaustive, and more than one register may be used to perform the action.

#	Action	Register usage
1	Reset the BCR-PLUS/BCR-LITE device or it's I2C port	RESET
2	Check if a device is attached on the Type-C port	TYPE_C_STATUS
3	Read the status of PD contract Check if port is DFP or UFP	PD_STATUS
4	Get the voltage and current requested by BCR-PLUS	CURRENT_PDO and CURRENT_RDO
5	Get changes to Type-C or PD port state	Poll: Read the EVENT_STATUS register for a list of changes since last read; write 1 to any bit to clear event Interrupt: Set events in EVENT_MASK and wait for interrupt. Then read PD_RESPONSE and clear interrupt.
6	Send a vendor defined message and read the device's response to the VDM	1: Set the 'VDM Received' bit in EVENT_MASK Use DM_CONTROL to send the VDM 2: Ensure PD_RESPONSE indicates SUCCESS 3: Wait for 'VDM Received' event in PD_RESPONSE and read Data Memory
7	Send a new PD request message and read the device's response	1: Get the Source PDOs list obtained from the port partner 2: Send the custom PD request message. 2: Ensure PD_RESPONSE indicates SUCCESS 3.a: If receives 'Accept Message Received' event in PD_RESPONSE and read Data Memory, then Wait for 'PS_RDY Message received' event in PD_RESPONSE and read Data Memory to ensure the new PD contract to be made. 3.b: If receives 'Reject Message Received' or 'Reject Message Received' event in PD_RESPONSE and read Data Memory, then it fails to establish the new PD contract.

HPI protocol

#	Action	Register usage
8	Disable legacy charging mode	CHARGING_MODE_DISABLE
9	Read the status of legacy charging mode	CHARGING_MODE_STATUS
10	Re-config the OCP settings	OCP_CONFIG
11	Re-config the OVP settings	OVP_CONFIG
12	Re-config the OTP settings	OTP_CONFIG
13	Read the voltage on VBUS line	BUS_VOLTAGE
14	Read the load current draw from the Bus line	BUS_CURRENT

3.9 DEVICE_MODE

This register indicates the active device mode. BCR-PLUS can be either in boot mode or application firmware mode. BCR-LITE always returns a constant byte from this register.

Name	DEVICE_MODE		
Address	0x0000		
Size	1-byte		
Field name	R/W	Field offset	Description
Current Mode	RO	Byte 0	b7-b2: 0b100000 b1-b0: Firmware mode 0 – Boot mode 1 – Application Firmware

3.10 SILICON_ID

The system controller can use this register to read the Silicon ID of the device. The Silicon ID is fixed for BCR-PLUS/BCR-LITE and will not change with the part.

This register can be used to identify if the HPI device is EZ-PD™ BCR-PLUS/EZ-PD™ BCR-LITE or any other device in the EZ-PD™ CCG family.

Name	SILICON_ID		
Address	0x0002		
Size	2-bytes		
Field name	R/W	Field offset	Description
Silicon ID	RO	Bytes [0..1]	BCR-PLUS device will always return 0x2011 when powered BCR-LITE device will always return 0x20A2 when powered

3.11 INTERRUPT

This register indicates the type of interrupt for which the INTR pin was asserted. The system controller writes to the register to clear the corresponding bit after reading corresponding response.

Name	INTERRUPT		
Address	0x0006		
Size	1-byte		
Field name	R/W	Field offset	Description
Interrupt Status	R/W1C	Byte 0	Bit 0: Device Interrupt 0 = No device interrupt pending 1 = New response available in DEV_RESPONSE register
			Bit 1: PD Port Interrupt 0 = No port interrupt pending 1 = New response available in PD_RESPONSE register
			Bits 7..2: Reserved (0x00)

3.12 PD_STATUS

This read-only status register reports the default PD configuration of BCR-PLUS and the current state of the PD contract.

Name	PD_STATUS		
Address	0x1008		
Size	4-bytes		
Field name	R/W	Field offset	Description
Default Config	RO	Byte 0 [5..0]	Bits 5..0: 0x00 (UFP Data Role and Sink Power Role) This field indicates the default configuration supported by BCR-PLUS
Current Config	RO	Byte 0 [7..6] Byte 1..3	Bit 6: Current Port Data Role 0 = UFP 1 = DFP This field is updated with the current data role when the Type-C port is attached to a partner device. The role can be changed by sending or receiving a DR_Swap message.
			Bit 7: Reserved (0)
			Bit 8: Current Port Power Role (0) 0 = Sink BCR-PLUS always functions as a sink and so this bit is never expected to be set to 1.
			Bit 9: Reserved (0)
			Bit 10: Contract State 0 = No contract exists with port partner 1 = Explicit PD contract exists with port partner “Current Config” bits (Bits 6..31) in this register are valid only if this bit is set to 1.

HPI protocol

Name	PD_STATUS		
Address	0x1008		
Size	4-bytes		
Field name	R/W	Field offset	Description
			Use the CURRENT_PDO and CURRENT_RDO registers to fetch details of the contract.
			Bits 13..11: Reserved (0)
			Bit 14: Sink Tx Ready Status (Rp state) 0 = Rp is in SinkTxOk (Sink can send messages to Source) 1 = Rp is in SinkTxNG (Sink shouldn't send messages)
			Bit 15: Policy Engine State (see Chapter 8 of PD spec) 0 = Port is not in PE_SNK_Ready state 1 = Port is in PE_SNK_Ready state
			Bits 17..16: PD Spec revision supported by BCR-PLUS 00 = PD 2.0 01 = PD 3.0 Others = Reserved
			Bit 18: Partner (attached device) PD Spec revision 0 = Partner is PD 2.0 device 1 = Partner is PD 3.0 device
			Bit 19: Partner un-chunked extended message support status 0 = Partner doesn't support un-chunked messages 1 = Partner supports un-chunked messages
			Bits 31..20: Reserved (0)

3.13 TYPE_C_STATUS

This read-only status register reports the status of the Type-C port on the BCR-PLUS/BCR-LITE device.

Name	TYPE_C_STATUS		
Address	0x100C		
Size	4-bytes		
Field name	R/W	Field offset	Description
Current Type-C Status	RO	Byte 0	Bit 0: Port Partner Connection Status 0 = Port is not connected to partner 1 = Port is connected to partner Other bits in this register are valid only if this bit is set.
			Bit 1: CC Polarity 0 = CC1 1 = CC2
			Bits 4..2: Attached device type 000 = Nothing attached 010 = Source attached 011 = Debug Accessory attached <i>Other values = Reserved</i>

HPI protocol

Name	TYPE_C_STATUS		
Address	0x100C		
Size	4-bytes		
Field name	R/W	Field offset	Description
			Bit 5: Reserved (0)
			Bits 7..6: Type-C Current Level (Rp of partner device) 00 = Default (900mA) 01 = 1.5A 10 = 3A 11 = Reserved This is the Rp value advertised by the attached port partner. If the CC line voltage is invalid, then this field will indicate 00.

3.14 BUS_VOLTAGE

This read-only status register reports the live voltage on the VBUS supply for the specified port (in 100-mV). The voltage will be measured using the internal ADC on the BCR-PLUS/BCR-LITE device at the time of reading this register.

Name	BUS_VOLTAGE		
Address	0x100D		
Size	1-byte		
Field name	R/W	Field offset	Description
Voltage on the VBUS line	RO	Byte 0	Voltage Live VBUS voltage, in 100 mV

3.15 BUS_CURRENT

This read-only status register reports the live current draw from the VBUS supply for the specified port (in 50-mA). The current will be measured using the internal ADC and external resistor (5 mΩ).

Name	BUS_CURRENT		
Address	0x1058		
Size	1-byte		
Field name	R/W	Field offset	Description
Current draw from the bus supply	RO	Byte 0	Current Live VBUS current, in 50 mA

3.16 CURRENT_PDO

This read-only status register holds the active PDO requested by BCR-PLUS from the attached power adapter. This field has the voltage(s) and maximum current (or power) advertised by the attached power adapter.

Refer to the [Power Delivery specification](#) for details on how to interpret the bits in this field.

Name	CURRENT_PDO		
Address	0x1010		
Size	4-bytes		
Field name	R/W	Field offset	Description
Current PDO	RO	Bytes 0..3	Active PDO This is the 32-bit Power Data Object (PDO) that was most recently received from the attached source. Refer to Chapter 6.4 of the PD 2.0 or 3.0 specifications for details on how to interpret the bits in a PDO.

3.17 CURRENT_RDO

This read-only status register reports the Request Data Object (RDO), which was used to establish the PD contract. This register contains the maximum operating current and operating current values requested by BCR-PLUS.

Name	CURRENT_RDO		
Address	0x1014		
Size	4-bytes		
Field name	R/W	Field offset	Description
Current RDO	RO	Bytes 0..3	Active RDO. This is the 32-bit RDO that was most recently sent from BCR-PLUS to the attached Source. Refer to Chapter 6.4 of the PD 2.0 or 3.0 specifications for details on how to interpret the bits in an RDO.

HPI protocol

3.18 EVENT_STATUS

Instead of listening for events, the system controller can read this register at any time to know what has happened on the Type-C/PD port until then.

This is a “sticky” register, which means that any bit set to ‘1’ will remain so until the system controller clears it. The system controller must write ‘1’ to any bit to clear any event.

Name	EVENT_STATUS		
Address	0x1034		
Size	4-bytes		
Field name	R/W	Field offset	Description
Event Status	R/W1C	Bytes 0..3	Bit 0: Type-C Device Attached
			Bit 1: Type-C Device Disconnected
			Bit 2: PD Contract Negotiation Completed
			Bit 3: Power Role Swap Completed
			Bit 4: Data Role Swap Completed
			Bit 5: VCONN Swap Completed
			Bit 6: Hard Reset received
			Bit 7: Hard Reset sent
			Bit 8: Soft Reset sent
			Bit 9: Cable Reset sent
			Bit 10: Type-C Error Recovery initiated
			Bit 11: BCR-PLUS Entered Source Disabled state (i.e. Type-C-only charger detected)
			Bits 28..12: Reserved (0)
			Bit 29: Unexpected voltage on VBUS
			Bit 30: VBUS voltage is over expected voltage range
			Bit 31: VBUS current is over the limit value

3.19 CHARGING_MODE_STATUS

This read-only status register reports the active Legacy charging mode.

Name	CHARGING_MODE_STATUS		
Address	0x1095		
Size	1-bytes		
Field name	R/W	Field offset	Description
Charging Mode Status	RO	Bytes 0	Active Legacy Charging Mode 0x00 – no legacy charging mode 0x01 – BC 1.2 DCP/CDP charging 0x02 – QC2.0 charging 0x03 – Reserved 0x04 – AFC charging 0x05 – Apple charging

Command registers

4 Command registers

Command Registers used to trigger changes in the BCR-PLUS/BCR-LITE device or in the Type-C (or PD) port state. Any write to a Command register will result in the INTR pin being asserted and a response populated in one of the Response registers.

After writing to a Command register, the system controller must wait for the INTR pin to assert. The time taken to assert is dependent on the command being executed.

Once the response is read (from either [DEV_RESPONSE](#) or [PD_RESPONSE](#)), the system controller must clear the interrupt by writing to the corresponding bit in [INTERRUPT](#).

4.1 RESET

This register is used to either reset the device or the I2C block.

When the I2C block is reset, the I2C module used for the HPI is reconfigured. All outstanding commands and responses are flushed.

When the device is reset, BCR-PLUS/BCR-LITE undergoes a full chip reset causing the Power Delivery contract to be renegotiated and the sink FET switch to be opened.

Note: This will result in the system losing power if no alternative power source (such as a battery) is available.

Name	RESET		
Address	0x0008		
Size	2-bytes		
Field name	R/W	Field offset	Description
Signature	WO	Byte 0	“R” (0x52) A reset is executed only if the signature byte is ASCII ‘R’ (0x52)
Reset type	WO	Byte 1	Bit 0: Reset Type 0 = I2C Reset 1 = Device Reset
			Bits 7..1: Reserved (0x00)

Responses in DEV_RESPONSE after writing to this register:

- SUCCESS (0x02):** When the I2C block has been reset successfully
- RESET_COMPLETE (0x80):** When the device has been reset successfully
- INVALID_COMMAND (0x09):** When the signature is not ‘R’, or if any of the reserved bits are not ‘0’

4.2 EVENT_MASK

This register will enable the set of events that BCR-PLUS will notify the system controller about.

Whenever BCR-PLUS detects an event (or PD message) for which a mask bit is set, it asserts the INTR pin and sets the “Port Interrupt” bit in [INTERRUPT](#) register.

The actual event code will be available in the [PD_RESPONSE](#) register and any associated data will be available in the data memory.

Upon device reset, all mask bits are set to ‘0’.

Command registers

Responses in DEV_RESPONSE after writing to this register:

SUCCESS (0x02): Any write to [EVENT_MASK](#) will result in this response.

Name	EVENT_MASK		
Address	0x1024		
Size	4-bytes		
Field name	R/W	Field offset	Description
Event filter	R/W	Bytes 0..3	Bit 0: Reserved (0)
			Bit 1: Overcurrent detected event
			Bit 2: Overvoltage detected event
			Bit 3: Type-C device connected event
			Bit 4: Type-C device disconnected event
			Bit 5: PD contract negotiation completed event
			Bit 6: PD control message received event
			This event mask bit will enable the following events:
			- Data Role Swap completed
			- PS_RDY Message received
			- Accept Message received
			- Reject Message received
			- Wait Message received
			- Hard Reset received
			Bit 7: Vendor-defined message (VDM) received event
			Bit 8: Source capabilities message received
			Bit 9: Sink capabilities message received
			Bit 10: Reserved
			Bit 11: Error and timeout events
			This event mask bit enables the following events
			- Hard Reset sent
			- Soft Reset sent
			- Source Disabled state entered (PD not detected)
			- Sender ResponseTimer timeout
			- No response received for VDM
			- Unexpected voltage on VBUS
			- Type-C Error Recovery
			Bit 12: Reserved (0)
			Bit 13: Rp value change detected event
			Bits 15..14: Reserved (0)
			Bit 16: Overtemperature event
			Bit 17: PD 3.0 extended data message received event
			Bits 31..18: Reserved (0)

Command registers

4.3 DM_CONTROL (Send PD Data Message)

The system controller can use this register to send various Power Delivery Data Messages to the Port Partner (SOP) or Type-C cable marker (SOP' or SOP'') devices.

This register is most typically used to send vendor defined messages (VDM) to the port partner to perform design-specific operations.

To send any message, the system controller follows the steps:

1. Update the data memory with the complete PD 4data message (excluding the Message Header, which will be prepended by BCR-PLUS). The message should be written in 32-bit chunks in little endian order. When sending VDMs, the data memory must also contain the VDM header along with the rest of the actual message.
2. Write to the DM_CONTROL register with the recipient Type (SOP=Port Partner, SOP*=Cable) and the size of the Data Message in bytes.
3. BCR-PLUS will attempt to send the message. If the message was sent properly or if any errors are seen, it will notify the system controller and the response code will be placed in PD_RESPONSE register.

If the port partner sends any response to the message sent by BCR-PLUS, then BCR-PLUS will notify the system controller separately. To listen for such partner messages, write to the corresponding bit in the [EVENT_MASK](#) register or poll the [EVENT_STATUS](#) register.

NAME ADDRESS SIZE	DM_CONTROL 0x1000 2-bytes		
Field name	R/W	Field offset	Description
PD Data Message Send Settings	WO	Bytes 0..1	Bits 1..0: Packet recipient type 00 = SOP 01 = SOP' (Near End of Cable) 10 = SOP'' (Far End of Cable) Others = Reserved
			Bit 2: PD 3.0 data message Set this bit when sending alert or Battery_Status messages
			Bit 3: Extended Data Message Set this bit when sending PD 3.0 Extended Data Messages
			Bit 4: Disable <i>SenderResponseTimer</i> Set this bit to not use the SenderResponseTimer. Refer to Section 6.2.2 of the PD spec for details.
			Bit 5: Reserved (0)
			Bits 7..6: Message Length Bits 9..8 Bits 15..8: Message Length Bits 7..0 Length of the PD Data Message in bytes.

Responses in PD_RESPONSE after writing to this register:

1. **SUCCESS (0x02):** If the Data Message was transmitted successfully (i.e. GoodCRC was received from the recipient device)
2. **TRANSACTION_FAILED (0x0C):** If the transmission failed i.e. GoodCRC was not received

Command registers

3. **INVALID_COMMAND (0x09)**: If any reserved bits are set
4. **PD_COMMAND_FAILED (0x0D)**: If the Type-C port is not connected to a partner, or if a PD contract does not exist, or if the BCR-PLUS device is not ready to send the message. Check the “PE_Ready” and “Sink Tx OK” bits of PD_STATUS register to know if BCR-PLUS is ready to send a message.

4.4 SELECT_SINK_PDO

This register is used to selectively enable or replace the sink capabilities of the BCR-PLUS device. On power-up, BCR-PLUS has two sink capabilities (or Sink PDOs):

PDO #1: Fixed supply, 5V, 900mA, Higher_Capability = Yes, if VBUS_MAX > 5V else No

PDO #2: Variable supply, Minimum voltage = VBUS_MIN, Maximum_Voltage = VBUS_MAX, Operational Current = ISNK_COARSE + ISNK_FINE

These PDOs are configured-based on the resistor dividers on the four configuration pins on BCR-PLUS.

The system controller can change the Sink PDOs in the following two ways:

1. The system controller can enable/disable the two default PDOs. It does this by updating the PDO mask.
2. The default Sink PDOs can be replaced with a custom list. To do this, the system controller must update the Data Memory with the PDO list and write to the SELECT_SINK_PDO register for the new PDOs to take effect.

Note: Once the Sink PDOs are updated, BCR-PLUS will re-negotiate the PD contract with the power adapter. This will cause momentary power loss to the system.

To replace the default sink PDOs, the system controller must write the following packet into Data Memory and then write to the SELECT_SINK_PDO register.

1. Byte 0..3: ASCII string “SNKP” (i.e. 0x50 0x4B 0x4E 0x53)
2. Byte 4..7: Sink PDO #1
3. Byte 8..11: Sink PDO #2
4. Byte 12..15: Sink PDO #3
5. Byte 16..19: Sink PDO #4
6. Byte 20..23: Sink PDO #5
7. Byte 24..27: Sink PDO #6
8. Byte 28..31: Sink PDO #7

If the system controller wants to use fewer than seven PDOs, then it must set the remaining bytes to ‘0’.

Responses in PD_RESPONSE after writing to this register:

1. **SUCCESS (0x02)**: If the Sink PDOs are updated successfully.
2. **INVALID_ARGUMENT (0x09)**: If mask is ‘0’ or a non-existent PDO is enabled.

Command registers

Name	SELECT_SINK_PDO		
Address	0x1005		
Size	1-byte		
Field name	R/W	Field offset	Description
Sink PDO Mask	WO	Byte 0	Bit 0: Enable PDO 1 Bit 1: Enable PDO 2 Bit 2: Enable PDO 3 Bit 3: Enable PDO 4 Bit 4: Enable PDO 5 Bit 5: Enable PDO 6 Bit 6: Enable PDO 7 Bit 7: Set the “Unconstrained Power” bit in PDO 1 Once this register is written to, BCR-PLUS will check if the first 4 bytes of Data Memory has the “SNKP” signature. If signature is present, it updates the Sink PDO list and uses the mask as specified in Bits 6..0. If signature is not present, it enables PDOs selected by the mask in Bits 6..0. If all bits are 0x00 then BCR-PLUS will fall back to the default Sink PDOs as determined by the 4 configuration pins.

4.5 PD_CONTROL (Send PD control message)

This register is used to send Power Delivery control messages to the port partner.

The port partner may respond back to a PD Control message sent using this register. To look for these responses, poll the [EVENT_STATUS](#) register or enable the “PD Control Message Received” event in [EVENT_MASK](#) and wait for an interrupt from BCR-PLUS.

Name	PD_CONTROL		
Address	0x1006		
Size	1-byte		
Field name	R/W	Field offset	Description
Command	WO	Byte 0	0x00~0x09: Reserved
			0x0A: Send Get_Source_Cap message BCR-PLUS will attempt to retrieve the Port Partner’s Source Capabilities. Note: This will cause PD contract to be renegotiated resulting in a power loss to the system
			0x0B~0x0C: Reserved
			0x0D: Send Hard_Reset Note: This will cause PD contract to be renegotiated resulting in a power loss to the system
			0x0E: Send Soft_Reset message Note: This may cause PD contract to be renegotiated resulting in a power loss to the system
			0x0F~0x15: Reserved

Command registers

Name	PD_CONTROL		
Address	0x1006		
Size	1-byte		
Field name	R/W	Field offset	Description
			0x16: Send Get_Source_Cap_Extnd message
			0x17: Send Get_Status message
			0x18: Send Not_Supported message
			0x1C~0xFF: Reserved

Responses in PD_RESPONSE after writing to this register:

1. **SUCCESS (0x02)**: If the PD Control message was sent successfully i.e., GoodCRC was received from the port partner.
2. **TRANSACTION_FAILED (0x0C)**: If the transmission failed i.e., GoodCRC was not received.
3. **INVALID_COMMAND (0x09)**: If the command is any reserved value.
4. **PD_COMMAND_FAILED (0x0D)**: If the Type-C port is not connected to a partner, or if a PD contract does not exist, or if the BCR-PLUS device is not ready to send the message. Check the “PE_Ready” and “Sink Tx OK” bits of PD_STATUS register to know if BCR-PLUS is ready to send a message.

4.6 SINK_RDO_REQUEST

This register is used to send a custom Request Data Object (RDO) to the port partner.

A write to this register will take effect only if the Sink PDO mask (in SELECT_SINK_PDO) is non-zero. This requirement is made so that the system controller can fully determine the state of the PD contract.

When the system controller writes to this register, BCR-PLUS sends the request as-is to the port partner. BCR-PLUS will not validate the contents of the packet. If the packet is malformed or if some bits are not set correctly, then the port partner may issue Soft Resets or Hard Resets to BCR-PLUS causing the system to lose power.

If PD contract is established using PPS APDO, then BCR-PLUS will periodically request this PPS PRDO.

The port partner may respond back to a PD Control message sent using this register. To look for these responses, poll the [EVENT_STATUS](#) register or enable the “PD Control Message Received” event in [EVENT_MASK](#) and wait for an interrupt from BCR-PLUS.

Name	SINK_RDO_REQUEST		
Address	0x1054		
Size	4-bytes		
Field name	R/W	Field offset	Description
Request Data Object	WO	Byte 0..3	32-bit Request Data Object Active RDO Writing to this register will result in Request of PD contract re-negotiation, in little endian order. If PD contract is established using PPS APDO, then BCR-PLUS will periodically request this PPS PRDO.

Command registers

Responses in PD_RESPONSE after writing to this register:

1. **SUCCESS (0x02):** Request message was queued to be sent to the port partner. A SUCCESS does not guarantee that GoodCRC was received, nor does it guarantee successful PD Contract renegotiation. Look for events in the [EVENT_STATUS](#) register or listen to events using [EVENT_MASK](#) for more details.
2. **COMMAND_FAILED (0x06):** If the request message could not be sent then try again later.

4.6.1 PPS Request

The SINK_RDO_REQUEST register is used to initiate the request message from the sink port to the source port. BCR-PLUS does not check Request RDO value that is used in SINK_RDO_REQUEST register.

The system controller can read port partner Source PDOs, BUS_VOLTAGE and BUS_CURRENT (if it supported by board) and generate required RDO message.

When the system controller requests PPS Sink PRDO from SINK_RDO_REQUEST register the BCR-PLUS starts send request message with this RDO every 9s.

4.7 FAULT_PIN_MASK

This register sets a mask of fault conditions that report on the Fault pin. By default, all conditions are enabled.

The actual event code will be available in the [PD_RESPONSE](#) register and any associated data will be available in the data memory.

Name	FAULT_PIN_MASK		
Address	0x1080		
Size	1-bytes		
Field name	R/W	Field offset	Description
Fault pin mask	R/W	Byte 0	Bits 0: Mismatch event mask Setting this bit will enable either of the following events to assert FAULT pin: USB-PD contract exists with mismatch bit set (5 V contract but High_Cap bit is set. Source is not able to provide the power the sink needs). Legacy charging contract exists at 5 V although VBUS_MIN selects above 5 V. Source is not able to provide the power the sink needs.
			Bit 1: OVP event mask Setting this bit will enable the OVP event to assert FAULT pin.
			Bit 2: Sink OCP event mask Setting this bit will enable the sink OCP event to assert the FAULT pin.
			Bit 3: OTP event mask Setting this bit will enable the OTP event to assert the FAULT pin.
			Bits 7..4: Reserved

Command registers

Responses in PD_RESPONSE after writing to this register:

SUCCESS (0x02): Any write to FAULT_PIN_MASK will result in this response.

4.8 OVP_CONFIG

This register allows reconfiguration of over voltage protection by the system controller. To re-enable the OVP setting, you must set new threshold and debounce values.

This register is available only if OVP is enabled in the configuration table. If not, the BCR-PLUS/BCR-LITE returns “Not supported”.

Name	OVP_CONFIG		
Address	0x1088		
Size	4-bytes		
Field name	R/W	Field offset	Description
Signature	R/W	Byte 0	8-bits VBUS OVP command “D” (0x44) OVP Disable command is executed only if this signature byte is ASCII ‘D’ (0x44). “C” (0x43) Re-enable OVP and update threshold and debounce period if this byte is loaded by the signature: “C” (0x43). If signature is not valid, BCR-PLUS/BCR-LITE updates Response Register with ‘Invalid Command’ error code.
Threshold	R/W	Byte 1	8-bits VBUS OVP threshold OVP threshold set in percentage from contract VBUS voltage. max value of VBUS threshold is 40% (0x28).
Debounce	R/W	Byte 2	8-bits VBUS OVP threshold OVP debounce period in us between first interrupt and port disable.
Reserved	R/W	Byte 3	Bits 7..0: Reserved (0x00)

Responses in PD_RESPONSE after writing to this register:

- SUCCESS (0x02):** When the VBUS OVP command has been set successfully.
- Invalid Command (0x05):** If the signature is not valid, not ASCII ‘C’ or ASCII ‘D’.
- Not Supported (0x0A):** If BCR-PLUS/BCR-LITE does not enable the VBUS OVP protection in the configuration table.

Command registers

4.9 OCP_CONFIG

This register allows reconfiguration of over current protection by the system controller. To re-enable the OCP setting, you must set new threshold and debounce values.

This register is available only if OCP is enabled in the configuration table. If not, BCR-PLUS returns “Not Supported”.

Name Address Size	OCP_CONFIG 0x108C 4-bytes		
Field name	R/W	Field offset	Description
Signature	R/W	Byte 0	8-bits VBUS OCP command “C” (0x43) Re-enable OCP and update threshold and debounce period if this byte is loaded by the signature: “C”. “D” (0x44) OCP Disable command is executed only if this signature byte is ASCII ‘D’ (0x44). If the signature is not valid, BCR-PLUS updates the Response Register with an ‘Invalid Command’ error code.
Threshold	R/W	Byte 1	8-bits VBUS OCP threshold OCP threshold set in percentage from contract VBUS voltage. max value of OCP threshold is 40% (0x28).
Debounce	R/W	Byte 2	8-bits VBUS OCP Debounce OCP debounce period in ms between first interrupt and start fault condition
Retry Count	R/W	Byte 3	8-bits VBUS OCP retry count OCP retry count before port disable.

Responses in PD_RESPONSE after writing to this register:

1. **SUCCESS (0x02):** When the VBUS OCP command has been set successfully.
2. **Invalid Command (0x05):** If the signature is not valid, not ASCII ‘C’ or ASCII ‘D’.
3. **Not Supported (0x0A):** If BCR-PLUS does not enable the VBUS OCP protection in the configuration table.

Command registers

4.10 OTP_CONFIG

This register allows reconfiguration of over temperature protection by the system controller. To re-enable the OTP setting, you must set new threshold and debounce values.

This register is available only if OTP is enabled in the configuration table. If not, BCR-PLUS returns “Not Supported”.

Name	OTP_CONFIG		
Address	0x1090		
Size	4-bytes		
Field name	R/W	Field offset	Description
Signature	R/W	Byte 0	8-bits VBUS OTP command “D” (0x44) OTP Disable command is executed only if this signature byte is ASCII ‘D’ (0x44). “C” (0x43) Re-enable OTP and update threshold and debounce period if this byte is loaded by the signature: “C” (0x43). If signature is not valid, BCR-PLUS updates Response Register with ‘Invalid Command’ error code.
Threshold	R/W	Byte 1	8-bits VBUS OTP threshold OTP threshold set in max value of threshold is 100°C (0x64).
Debounce	R/W	Byte 2	8-bits VBUS OTP debounce OTP debounce period in ms between first interrupt and start fault condition (error recovery).
Hysteresis	R/W	Byte 3	Reserved

Responses in PD_RESPONSE after writing to this register:

1. **SUCCESS (0x02):** When the VBUS OTP command has been set successfully.
2. **Invalid Command (0x05):** If the signature is not valid, not ASCII ‘C’ or ASCII ‘D’.
3. **Not Supported (0x0A):** If BCR-PLUS does not enable the VBUS OTP protection in the configuration table.

Command registers

4.11 SNK_FET_DISABLE

This register allows you to disable sink consumer path if the USB-PD contract renegotiation operational current is equal to or less than 500mA (current $\leq$ 500mA), this register only applies to BCR-PLUS devices.

Name	SNK_FET_DISABLE		
Address	0x1084		
Size	1-bytes		
Field name	R/W	Field offset	Description
sink FET disable	R/W	Byte 0	DISABLE Disable SNK FET. Handle command only if this byte is loaded with valid signature by the system controller: “D” (0x44). RESET Return SNK FET to default state. Handle command only if this byte is loaded with valid signature by the system controller: “R” (0x52). If signature is not valid, BCR-PLUS updates the Response register with ‘Invalid Command’ error code.

Responses in PD_RESPONSE after writing to this register:

1. **SUCCESS (0x02):** When the sink FET disable command has been set successfully.
2. **Invalid Command (0x05):** If the signature is not valid, not ASCII ‘D’ or ASCII ‘R’.

4.12 SNK_FET_CONTROL

This register allows you to disable or enable the sink consumer path for BCR-PLUS/BCR-LITE.

Name	SNK_FET_CONTROL		
Address	0x1085		
Size	1-bytes		
Field name	R/W	Field offset	Description
Sink FET control	R/W	Byte 0	DISABLE Disable SNK FET. Handle command only if this byte is loaded with valid signature by the system controller: “D” (0x44). ENABLE Enable SNK FET. Handle command only if this byte is loaded with valid signature by EC: “E” (0x45). If signature is not valid, BCR-PLUS/BCR-LITE updates the Response register with ‘Invalid Command’ error code.

Responses in PD_RESPONSE after writing to this register:

1. **SUCCESS (0x02):** When the sink FET control command has been set successfully.
2. **Invalid Command (0x05):** If the signature is not valid, not ASCII ‘D’ or ASCII ‘E’.

Command registers

4.13 SAFE_PWR_CONTROL

This register allows you to disable or enable the sink safe power path for BCR-PLUS/BCR-LITE.

Name	SAFE_PWR_CONTROL		
Address	0x1086		
Size	1-bytes		
Field name	R/W	Field offset	Description
safe power enable pin control	R/W	Byte 0	DISABLE Disable SAFE_PWR_ENABLE pin behavior. Handle command only if this byte is loaded with valid signature by the system controller: “D” (0x44). ENABLE Enable SAFE_PWR_ENABLE pin behavior. Handle command only if this byte is loaded with valid signature by EC: “E” (0x45). If signature is not valid, BCR-PLUS/BCR-LITE updates the Response register with ‘Invalid Command’ error code.

Responses in PD_RESPONSE after writing to this register:

1. **SUCCESS (0x02):** When the sink safe power path control command has been set successfully.
2. **Invalid Command (0x05):** If the signature is not valid, not ASCII ‘D’ or ASCII ‘E’.

4.14 CHARGING_MODE_DISABLE

This register allows you to disable the legacy charging mode support for BCR-PLUS/BCR-LITE.

Name	CHARGING_MODE_DISABLE		
Address	0x1094		
Size	1-bytes		
Field name	R/W	Field offset	Description
Charging mode disable	R/W	Byte 0	8-bits Disable command 0x01 – Disable legacy charging.

Responses in PD_RESPONSE after writing to this register:

1. **SUCCESS (0x02):** When the charging mode disable command has been set successfully.
2. **Not Supported (0x0A):** If BCR-PLUS/BCR-LITE does not enable the legacy charging mode support in the configuration table.

Response registers

5 Response registers

5.1 DEV_RESPONSE

The Device Response (DEV_RESPONSE) register is used to store responses to commands sent on certain Command registers. In addition, this register also holds device events such as reset completion.

Name	DEV_RESPONSE		
Address	0x007E		
Size	2-bytes		
Field name	R/W	Field offset	Description
Response Code	RO	Byte 0	Bit 7: Type of response 0 = Response to a Command Register write 1 = Asynchronous Event generated by BCR-PLUS/BCR-LITE
			Bits 6..0: Response Code See Response Codes for a list of codes
Length	RO	Byte 1	Length of the response (in bytes) If this field is non-zero, the data memory will be filled with data up to this size.

5.2 PD_RESPONSE

The Port Response (PD_RESPONSE) register is used to store responses to commands sent on Command Registers that trigger port state changes or send PD messages.

This register also holds any PD message responses or asynchronous PD/Type-C events.

Name	PD_RESPONSE		
Address	0x1400		
Size	4-bytes		
Field name	R/W	Field offset	Description
Response Code	RO	Byte 0	Bit 7: Type of Response 0 = Response to a Command Register write 1 = Asynchronous event generated by BCR-PLUS/BCR-LITE
			Bits 6..0: Response Code See Response Codes for a list of codes
Length1	RO	Byte 1	Length of the response (in bytes) if length < 256 If this field is non-zero, the Data Memory will be filled with data up to this size.
Length2	RO	Byte 2..3	Length of the response (in bytes) if length ≥ 256 If this field is non-zero, the data memory will be filled with data up to this size. If length of the response is less than 256, then Length1 is the same as Length2. If length of the response is more than 256 bytes, then Length2 > Length1.

Response registers

5.3 Response codes

BCR-PLUS/BCR-LITE can send two types of responses:

1. Responses to Command registers

This can be of two types: responses to device-specific commands and responses to port-specific commands

2. Asynchronous events

This can also be of two types: events generated by device state change or events generated when PD port state changes

Hence, there are four tables for each type of response.

Note that any response code not listed here is reserved for future use. In addition, only asynchronous port events contain additional data, which the system controller can read. These are documented as an additional column in that table.

Type	Responses to commands	
Response name	Code	Description
No Response	0x00	No Response No outstanding command or event in BCR-PLUS/BCR-LITE. Or BCR-PLUS/BCR-LITE is processing a command that will take a long time to complete.
Success	0x02	Success Command was handled successfully. Refer to the specific Command Register definition to understand what a successful handling of command means.
Invalid Command or Argument	0x05, 0x09	Invalid Command or Argument Partial register writes, reserved bits set, unexpected command code or unexpected command sizes.
Not Supported	0x0A	Command Not Supported in mode Command is not supported in the current mode
Transaction Failed	0x0C	Transaction Failed The PD message was not sent successfully. GoodCRC was not received in response to BCR-PLUS sending the command. BCR-PLUS could not send the message due to a packet collision. Message was invalid or formatted incorrectly. If the system controller receives this response, it is expected to poll the “PE_Ready” and “SinkTxOk” bits in PD_STATUS and retry when these two bits indicate transmission readiness.
PD Command Failed	0x0D	PD Command Failed BCR-PLUS was unable to send the command because the port was not connected or if BCR-PLUS was handling another PD command. If the system controller receives this response, it is expected to poll the “PE_Ready” and “SinkTxOk” bits in PD_STATUS and retry when these two bits indicate transmission readiness.
Port Busy	0x12	PD Port Busy The command wasn't sent because the PD PHY was busy.

Response registers

Type	Asynchronous device events	
Response name	Code	Description
Reset Complete	0x80	Reset completed event BCR-PLUS/BCR-LITE completed a power-on reset or I2C initiated reset and is back in operational mode.
Message Queue Overflow	0x81	Message queue overflow The number of messages in the message queue exceeded capacity and further responses/events have been dropped. When the system controller detects this event, it must flush the queue by repeatedly clearing interrupts in the INTERRUPT register until there are no more interrupts.
Overcurrent detected on VBUS	0x82	VBUS current is outside expected range. When VBUS current is out of the current limit range, the OCP event is triggered. The Sink FET is turned off to protect the system from over-loaded.
Overvoltage detected on VBUS	0x83	VBUS voltage is outside expected range. When VBUS is out of the (VBUS_MIN, VBUS_MAX) range, the OVP event is triggered. The Sink FET is turned off to protect the system from excess voltage.

Type	Asynchronous port events		
Response name	Code	Description	Data sent with response
Type-C Port Connected	0x84	Type-C attach detected. A new device was attached. BCR-PLUS/BCR-LITE detected the device and entered the Attached.SNK state in the Type-C Sink state machine.	None
Type-C Port Disconnected	0x85	Type-C detach detected. Device was detached and BCR-PLUS/BCR-LITE entered the Unattached.SNK state.	None
PD Contract Negotiation Complete	0x86	PD contract negotiation complete BCR-PLUS finished the process of negotiating a power contract with the attached adapter. The contract may have succeeded or failed. The result is available in the 8-bytes of response data sent along with this response.	8-bytes in data memory Bit 0: Set if PD contract negotiation is successful Bit 1: Set if BCR-PLUS sets "Capability Mismatch" in the Request message. Bit 2..4: Reason for PD Contract Negotiation Failure: 011 = Power adapter rejected BCR-PLUS's request and there was a previous explicit PD contract 100 = Power adapter rejected BCR-PLUS's request and there was no

Response registers

Type	Asynchronous port events		
Response name	Code	Description	Data sent with response
			explicit PD contract 101 = Power adapter accepted BCR-PLUS's request but didn't send PS_RDY Others = Reserved Bits 7..5: Reserved (0) Bytes 1..3: Reserved (0) Bytes 4..7: Request Data Object sent by BCR-PLUS to Power Adapter
Swap Completed	0x87	PD swap completed. A data role swap or VCONN direction swap was completed. The process may have succeeded or failed. Read the response data in data memory for more details.	8-bytes in data memory Bit 0: Set if PD contract negotiation is successful. Bit 1: Set if BCR-PLUS sets "Capability Mismatch" in the Request message. Bit 2..4: Reason for PD Contract Negotiation Failure: 011 = Power adapter rejected BCR-PLUS's request and there was a previous explicit PD contract 100 = Power adapter rejected BCR-PLUS's request and there was no explicit PD contract 101 = Power adapter accepted BCR-PLUS's request but didn't send PS_RDY Others = Reserved Bits 7..5: Reserved (0) Bytes 1..3: Reserved (0) Bytes 4..7: Request Data Object sent by BCR-PLUS to Power Adapter
PS_RDY	0x8A	PS_RDY message received	None
GotoMin	0x8B	GotoMin message received. Power adapter requested the system to enter the lower current mode	None
Accept	0x8C	Accept message received	None
Reject	0x8D	Reject message received	None
Wait	0x8E	Wait message received	None
Hard_Reset	0x8F	Hard reset received	None
VDM Received	0x90	PD VDM (Vendor Defined Message) Received	Length = 4 + 4 * Number_of_VDOs Byte 1..0: Message Header Byte 2: SOP type (0=SOP, 1=SOP',

Response registers

Type	Asynchronous port events		
Response name	Code	Description	Data sent with response
		BCR-PLUS responds automatically to certain Structured VDMs defined in the PD spec. The system controller can respond to Unstructured VDMs, and commands associated with alternate modes if necessary. Note that if a response will be sent, it must be done within tVDMsenderResponse time as defined in the PD spec. The data memory has the actual VDM message received from the port partner.	2=SOP") Byte 3: Reserved (0) Byte 4..7: VDM header Byte 8..N: VDM data
Source Capabilities	0x91	Source capabilities received. The port partner sent a new set of Source Capabilities message. This will cause the PD contract to be re-negotiated.	Length = 4 + 4 * Number_of_PDOs Byte 1..0: Message Header Byte 2: SOP type (0) Byte 3: Reserved (0) Byte 4..N: Source PDOs received from port partner
Sink Capabilities	0x92	Sink capabilities received Port partner sent sink capabilities message.	Length = 4 + 4 * Number_of_PDOs Byte 1..0: Message Header Byte 2: SOP type (0) Byte 3: Reserved (0) Byte 4..N: Sink PDOs received from port partner
Hard Reset Sent	0x9A	Hard reset was sent to port partner BCR-PLUS sends Hard Resets in response to certain events. Refer to the PD spec for details.	None
Soft Reset Sent	0x9B	Soft Reset was sent to port partner BCR-PLUS sends Soft Resets when certain Atomic Message Sequences are interrupted or under certain other conditions. Refer to the PD spec for details.	None
Source Disabled	0x9D	<i>SourceDisabled</i> state entered BCR-PLUS enters this state if it doesn't detect Power Delivery messages from the power adapter. The attached adapter may be Type-C only or a legacy charger.	None
No VDM Response Received	0x9F	No VDM Response Received	None

Response registers

Type	Asynchronous port events		
Response name	Code	Description	Data sent with response
		BCR-PLUS reports this error when a VDM was sent by system controller, but port partner did not send a response back to BCR-PLUS.	
Type-C Error Recovery	0xA1	Type-C Error Recovery BCR-PLUS/BCR-LITE entered the Type-C Error Recovery State. Refer to the Type-C Connector Specification for details on this state.	None
Battery Status Received	0xA2	Battery_Status message received	None
Alert Received	0xA3	Alert message received	None
Not Supported received	0xA4	Not_Supported message received	None
Rp change detected	0xAA	Rp change detected	None
Extended Data Message Received	0xAC	PD 3.0 Extended Data Message Received	Length = 4 + Extended message length Byte 1..0: Message Header Byte 3..2: Extended Message Header Byte 4..N: Data from the extended message
CC OVP	0xBA	CC overvoltage detected Voltage on CC exceeded expected range. This may be due to CC shorting to VBUS.	None

BCR-PLUS/BCR-LITE has a 512-byte data memory region split into two portions, for writes and reads. Any command that requires data to be sent or received will use the data memory region.

5.4 Data Memory Regions

The Data Memory is split into two regions: Read Data Memory and Write Data Memory.

Data memory region	Access	Address range	Description
Read Data Memory	RO	0x1404 to 0x150B	The 264-byte Read Data Memory is placed adjacent to the PD_RESPONSE register so that it can be read in one single I2C read. This region will contain the PD Message data, contract information and other information as listed in the Response Codes section.
Write Data Memory	WO	0x1800 to 0x19FF	The 512-byte Write Data Memory is used when writing to Command Registers to send additional data for the command.

Any write or read from a Data Memory region will not trigger an INTR assertion. Reads or writes beyond the expected address range will be NAKed.

Revision history**Revision history**

Document revision	Date	Description of changes
**	2021-08-31	New BCR-PLUS/BCR-LITE HPI Guide
*A	2023-10-03	Removed “restricted”. Updated the part number typo in the header to “CYPD3178”.

Trademarks

All referenced product or service names and trademarks are the property of their respective owners.

Edition 2023-10-03

Published by

Infineon Technologies AG

81726 Munich, Germany

**© 2023 Infineon Technologies AG.
All Rights Reserved.**

Do you have a question about this document?

Go to erratum@infineon.com

Document reference

002-33575 Rev. *A

Important notice

The information given in this document shall in no event be regarded as a guarantee of conditions or characteristics ("Beschaffheitsgarantie").

With respect to any examples, hints or any typical values stated herein and/or any information regarding the application of the product, Infineon Technologies hereby disclaims any and all warranties and liabilities of any kind, including without limitation warranties of non-infringement of intellectual property rights of any third party.

In addition, any information given in this document is subject to customer's compliance with its obligations stated in this document and any applicable legal requirements, norms and standards concerning customer's products and any use of the product of Infineon Technologies in customer's applications.

The data contained in this document is exclusively intended for technically trained staff. It is the responsibility of customer's technical departments to evaluate the suitability of the product for the intended application and the completeness of the product information given in this document with respect to such application.

Warnings

Due to technical requirements products may contain dangerous substances. For information on the types in question please contact your nearest Infineon Technologies office.

Except as otherwise explicitly approved by Infineon Technologies in a written document signed by authorized representatives of Infineon Technologies, Infineon Technologies' products may not be used in any applications where a failure of the product or any consequences of the use thereof can reasonably be expected to result in personal injury.